
E-PAAS

Electronic Product & Claim Approval Application System **Technical Design Document**

Prepared for: FSSAI (Food Safety and Standards Authority of India)
Version: 1.0 | Date: April 2026 | Status: Draft

Version History

Version	Author	Date	Notes
1.0	Development Team	April 2026	Initial draft — based on E-PAAS SRS v1.1

1. Project Overview

E-PAAS (Electronic Product and Claim Approval Application System) is a government-grade web platform built for FSSAI — India's national food safety regulator. It digitises the complete lifecycle of product approval applications submitted by applicants.

Before E-PAAS, applicants had to submit paper applications and physically follow up with authority. E-PAAS makes the entire process online: applicants apply digitally, authorised officers review through a structured multi-level workflow, and formal decisions are communicated back electronically with a full audit trail.

1.1 Application Types Covered

Code	Full Name	What It Covers
NSF	Non-Specified Food & Food Ingredients	Products with no existing legal standard in India. Cannot be sold without FSSAI approval.
CA	Claim Approval	Health or disease-related label claims. Requires scientific evidence and FSSAI approval.
AA	Ayurveda Aahara	Ayurvedic food products. Four sub-categories (A, B, B1, B2).
rPET	Recycled PET Packaging	Companies using recycled plastic for food packaging. Requires site inspection and FSSAI authorization (not just approval).
Any Other	Any Other Mandated Category	Any product type where FSS Regulations explicitly require prior FSSAI approval.

1.2 Core Features

Feature	Description	SRS Reference
Role-based dashboards	Every user sees only their own workspace, queues, and actions	FR-001
Application submission	Multi-step form with document upload, fee calculation, and payment	FR-002
Assignment & scrutiny	Nodal Officer assigns cases; Technical Officer reviews documents	FR-003
Query management	Back-and-forth clarification with 30-day response deadlines	FR-004
Expert Committee review	EC evaluates and records decision; Nodal Point B uploads minutes	FR-005
Decision letters	Formal approval/rejection letters generated and dispatched	FR-006
Appeal management	Applicant challenges rejection; CEO decides	FR-007

Review management	Final escalation after appeal rejection; Chairperson is the last authority	FR-008
Time extension requests	Applicant requests more time to respond to queries	FR-009
Reports & search	Ageing buckets, status sheets, full lifecycle search console	FR-010 to FR-012
Admin monitoring	Read-only visibility of all applications and current officer	FR-016

2. System Architecture

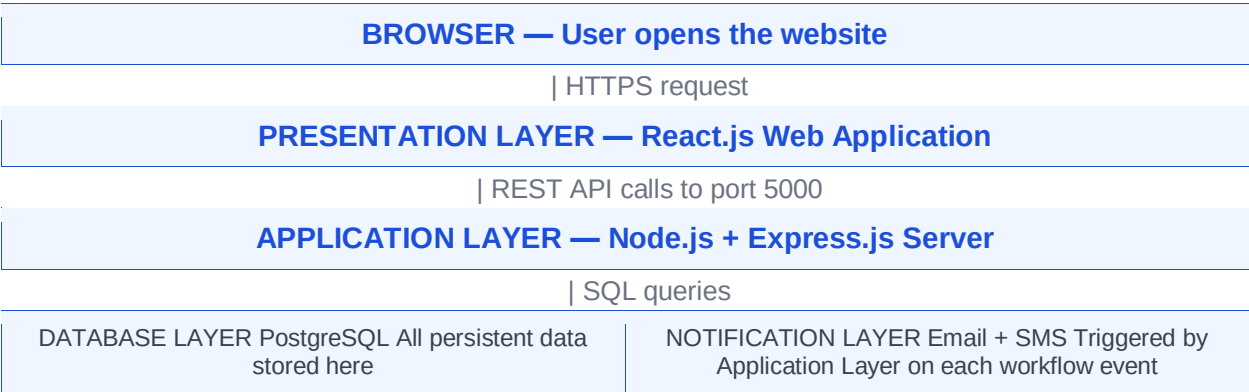
E-PAAS uses a four-layer architecture as defined in SRS Section 2.2. Each layer is physically separated into its own folder and communicates with adjacent layers through defined interfaces only.

2.1 The Four Layers

Layer	Technology	What It Does
Presentation Layer	React.js + Tailwind CSS + Shadcn/ui	The web application users open in their browser. Communicates with the backend via REST API calls only.
Application Layer	Node.js + Express.js	The backend server. Contains all business logic, workflow rules, role checks, API endpoints, and file handling.
Database Layer	PostgreSQL 16	Stores all data — applications, users, decisions, documents, workflow history, notifications.
Notification Layer	Nodemailer + MSG91/Twilio	Standalone service that sends Email and SMS alerts whenever a workflow event occurs.

2.2 Architecture Flow Diagram

| Read from top to bottom. Each arrow shows where data flows.



2.3 Technology Stack

Layer	Technology	Version	Purpose
Frontend	React.js	18.x	User interface

Frontend Styling	Tailwind CSS + shadcn/ui	3.x	Component library and styling
Frontend Routing	React Router	v6	Client-side navigation
Frontend State	React Context / Zustand	—	Auth state, logged-in user role
Backend Runtime	Node.js	20.x LTS	Server execution environment
Backend Framework	Express.js	4.x	REST API routing and middleware
Database	PostgreSQL	16.x	Primary data storage
Authentication	JWT (jsonwebtoken)	—	Stateless session management
Password Security	bcryptjs	—	Password hashing — never stored as plain text
File Storage	AWS S3 / Local	—	Uploaded application documents
Email	Nodemailer	—	Transactional email notifications
SMS	MSG91 / Twilio	—	SMS alerts to applicants and officers
Module Format	CommonJS (backend)	—	require() / module.exports for Node.js
Module Format	ESM (frontend)	—	import / export for React.js

3. User Roles and Permissions

E-PAAS has 8 distinct roles. The entire system adapts based on who is logged in — dashboard, menus, data, and available actions all change per role. This is enforced at the API level through role middleware, not just on the frontend.

Important: Multiple officers can hold the same role. There can be many Nodal Officers, Technical Officers, Expert Committee members, and Admins. View-only access to monitor applications and identify which officer / level the application is currently on. No workflow action shall be available.

3.1 Role Summary

Role	Core Job	Multiple People?
Applicant	Submits applications, responds to queries, files appeals and reviews, tracks status	Yes
Nodal Officer A	Receives applications, assigns Technical Officers, approves queries, dispatches final decisions	Yes
Technical Officer	Reviews documents, performs scrutiny, drafts queries, recommends EC, prepares decision letters	Yes
Expert Committee	Evaluates referred applications, records Approved/Rejected/Clarification decision	Yes
Nodal Point B	Uploads the final Decision	Yes
CEO	Appellate authority when applicant files an appeal after rejection	No
Chairperson	Final review authority . Chairperson's decision is absolute final.	No
Admin	View-only access to monitor applications and identify which officer / level the application is currently on. No workflow action shall be available.	No

3.2 Permission Matrix

Action	App	NA	TO	EC	NB	CEO	Chr	Adm
Submit application	Y	N	N	N	N	N	N	N
Respond to query	Y	N	N	N	N	N	N	N
File appeal	Y	N	N	N	N	N	N	N
File review (after appeal rejection)	Y	N	N	N	N	N	N	N
Request time extension	Y	N	N	N	N	N	N	N
Assign officer to application	N	Y	N	N	N	N	N	Y

Approve and dispatch query	N	Y	N	N	N	N	N	N
Draft query for applicant	N	N	Y	N	N	N	N	N
Perform document scrutiny	N	N	Y	N	N	N	N	N
Recommend application to EC	N	N	Y	N	N	N	N	N
Prepare decision letter	N	N	Y	N	N	N	N	N
Record EC decision outcome	N	N	N	Y	N	N	N	N
Upload EC approved decision	N	N	N	N	Y	N	N	N
Dispatch decision letter to applicant	N	Y	N	N	N	N	N	N
Appeal Decision	N	N	N	N	N	Y	N	N
Final review decision (Chairperson)	N	N	N	N	N	N	Y	N
View all applications (read-only)	N	N	N	N	N	N	N	Y
Create / manage officers accounts	N	N	N	N	N	N	N	Y

App=Applicant NO=Nodal Officer A TO=Technical Officer EC=Expert Committee NB=Nodal Point B
Chr=Chairperson Adm=Admin

4. Database Tables (Data Model)

All data is stored in PostgreSQL. Each table below is like a spreadsheet — every row is one record, every column is one piece of information. Tables are linked to each other through foreign keys (FK) — a value in one table that points to a row in another table.

4.1 users

Every person with a login — applicants, all types of officers, CEO, Chairperson, Admin.

users	
id PK	Unique identifier (string) for this user
role FK → roles	References the roles table — links user to their role
username	Login username
email	Login email address (unique across all users)
office_location	For Applicants, office location is not required. Since they are not assigned to any FSSAI location.
is_active	true/false — whether the account is enabled
created_at	Timestamp when account was created

4.2 roles (new table)

Lookup table for all roles in the system. Role is now a foreign key relationship instead of a plain string stored in users.

roles	
id PK	Unique role ID
role_code	Short code: Applicant NodalOfficerA TechnicalOfficer ExpertCommittee NodalPointB CEO Chairperson Admin
role_name	Human-readable name (e.g. Nodal Officer A)
description	Description of the role's responsibilities

4.3 applications ← Central Table

The heart of the system. Every product approval application is one row here. Every other table links back to this one.

applications	
id PK	Unique ID (string)
application_ref_no	Auto-generated reference shown to users (e.g. 2026-04-NSF-0001)
application_type	NSF CA AA rPET AnyOther

applicant_name	Name of the applicant / company (renamed from company_name)
address	Applicant address
product_name	Name of the product being applied for
food_category	FoSCoS food category code
kind_of_business	Manufacturer Relabeller Importer
current_stage	Where the application is RIGHT NOW in the workflow
ec_status	Expert Committee outcome: Approved Rejected Clarification Pending
final_status	Overall result: Approved Rejected Withdrawn Pending
submitted_date	When submitted — renamed from submitted_at
created_at	When the draft was first started
updated_at	Last modification timestamp

4.4 fee_payments (new table)

Payment records for each application. Payment fields have been moved out of the applications table into this dedicated table, allowing multiple payment records per application if needed.

fee_payments	
id PK	Unique payment record ID
application_id FK → applications	Which application this payment belongs to
category	Fee category (e.g. NSF, AA-A, CA)
fee_type	Type of fee (e.g. Application Fee, Processing Fee)
amount	Amount in INR (decimal)
payment_status	Paid Pending Waived
payment_date	Timestamp when payment was made
transaction_ref	Payment gateway transaction reference number

4.5 documents (was: application_documents)

References to every file uploaded with an application. Actual files live in S3 or local storage; this table stores the metadata and link.

documents	
id PK	Unique document ID
application_id FK → applications	Which application this belongs to

document_type	Label: Test Report Product Label Composition Sheet EC Minutes etc.
file_ref	Reference/URL to the file in storage — renamed from file_path
uploaded_by FK → users	Who uploaded it — renamed from uploaded_by_user_id
uploaded_at	Upload timestamp

4.6 workflow_actions ← Audit Trail

Every single move made on any application is permanently recorded here. This is the complete audit trail — who did what, when, and why.

workflow_actions	
id PK	Unique action ID
application_id FK → applications	Which application was acted on
from_role	Role from which the action originated
to_role	Role or stage the application moved to
action_type	Assign Query Response ForwardToEC Decision Appeal Review Extension Withdraw Dispatch
action_date	Exact date and time of the action
remarks	Notes written by the officer
supporting_document_ref	Reference to any document uploaded with this action — renamed from supporting_doc_ref

4.7 queries

When a Technical Officer needs clarification from the applicant, they raise a query. This table tracks the full lifecycle of each query.

queries	
id PK	Unique query ID
application_id FK → applications	Which application
drafted_by FK → users	Technical Officer who drafted the query — renamed from drafted_by_officer_id
approved_by FK → users	Nodal Officer A who approved and dispatched it — renamed from approved_by_nodal_id
query_text	The actual question / clarification being asked
response_text	Applicant's reply
response_date	When applicant submitted their response — renamed from response_submitted_at

status	Draft Sent Responded Closed
due_date	Response deadline (sent date + 30 days) — renamed from response_due_date

4.8 decisions

Formal decision records from Expert Committee, CEO (appeal), and Chairperson (review).

decisions	
id PK	Unique decision ID
application_id FK → applications	Which application
decision_stage	EC Appeal Review
decision_type	Approved Rejected Clarification Authorized NotAuthorized
decision_authority	ExpertCommittee CEO Chairperson — renamed from decision_authority_role
decision_date	Date the decision was made
letter_ref	Reference number of the formal communication letter

4.9 appeals

When an application is rejected, the applicant has the right to appeal. This table tracks the appeal and the authority's decision.

appeals	
id PK	Unique appeal ID
application_id FK → applications	Which application is being appealed
appeal_type	Type of appeal (new field from ER diagram)
filed_date	When the appeal was filed — renamed from filed_at
decided_by FK → users	The authority who decided on the appeal (CEO) — replaces filed_by_applicant_id
decision	Approved Rejected
decision_date	When the decision was made — renamed from decided_at
remarks	Decision reasoning — replaces ceo_remarks

4.10 extension_requests (was: time_extension_requests)

Applicants can request more time to respond to a query if the standard deadline is not sufficient.

extension_requests	
id PK	Unique ID
application_id FK → applications	Which application
requested_by FK → users	Who requested the extension — renamed from applicant_id
requested_date	The new deadline being requested
reason	Reason for needing more time — renamed from applicant_remarks
status	Pending Approved Rejected
approved_by FK → users	Officer who approved or rejected — renamed from actioned_by

4.11 notifications

Every workflow event generates a notification stored here for in-app display. Emails and SMS are sent separately via the Notification Layer.

notifications	
id PK	Unique notification ID
user_id FK → users	Who should receive this notification
application_id FK → applications	Related application
message	Notification text
is_read	Whether the user has read it (true/false)
created_at	When created — renamed from sent_at

4.12 workflow_templates (new table)

Defines and versions the workflow configuration for each application type. Controls which stages are valid start and end points and when each version is active.

workflow_templates	
id PK	Unique template ID
template_code	Business code identifying this workflow template
application_type	Application type this template applies to (NSF CA AA rPET AnyOther)
version	Version number — allows workflow rules to be updated without affecting existing applications
is_start_stage	true/false — flags the entry stage of this workflow
is_end_stage	true/false — flags a valid closure stage
effective_from	When this version became active

effective_to	When this version was superseded (null if currently active)
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4.13 reviews

After CEO rejects an appeal, the applicant may escalate to the Chairperson for a final review. This is the absolute last step — no further escalation is possible after the Chairperson decides.

reviews	
review_id PK	Unique review ID
application_id FK → applications	Which application
appeal_id FK → appeals	The rejected appeal this review is based on
filed_by_applicant_id FK → users	Applicant who filed the review
review_reason	Written grounds for the review petition
supporting_docs	Path to uploaded supporting documents
filed_at	When the review was filed
chairperson_decision	Approved Rejected — THIS IS FINAL. No further escalation.
decided_at	When the Chairperson made the decision

4.14 approval_numbers

When an application is finally approved, a structured approval number is auto-generated and permanently recorded here.

approval_numbers	
approval_id PK	Unique ID
application_id FK → applications	Which application was approved
approval_number	Structured number (e.g. 2026-APP-FSSAI-NSF-00001)
issued_date	Date of approval
issued_by_user_id FK → users	Nodal Officer who issued the approval
valid_until	Expiry date if applicable

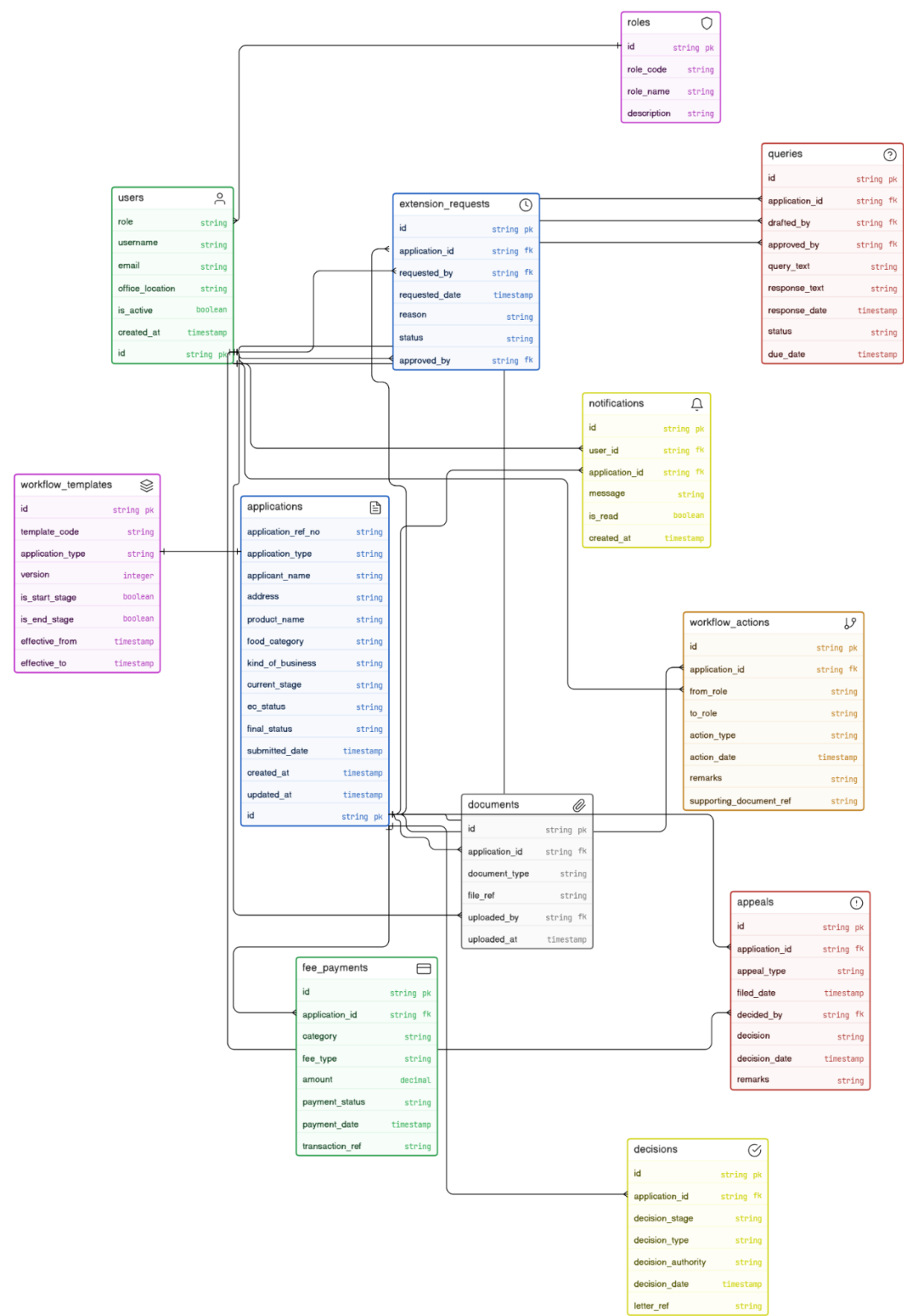
5. Entity Relationship Diagram

The diagram below shows how all database tables connect to each other. An arrow from Table A to Table B means A has a foreign key pointing to B (A 'belongs to' B).

5.1 Relationship Summary

Relationship	Type	Description
users → applications	One-to-Many	One applicant can submit many applications over time
users → applicant_profiles	One-to-One	Each applicant has exactly one profile with company details
applications → application_documents	One-to-Many	One application can have many uploaded files
applications → workflow_actions	One-to-Many	Every action taken on an application is permanently logged here
applications → queries	One-to-Many	An application can go through multiple rounds of queries
applications → decisions	One-to-Many	Up to 3 decisions: EC decision, CEO appeal decision, Chairperson review decision
applications → appeals	One-to-One	At most one appeal per application
appeals → reviews	One-to-One	At most one review per appeal
applications → approval_numbers	One-to-One	One approval number per approved application
queries → time_extension_requests	One-to-Many	Multiple extension requests can be made for a query
users → notifications	One-to-Many	Every user can receive many notifications
officer_assignments → users	Many-to-One	Many assignments can point to one officer

5.2 ER diagram Schema



6. Workflow Engine

The workflow engine is the core of E-PAAS. It controls which officer sees an application, what actions they can take, and where the application moves next. Every stage transition: (1) updates current_stage in the applications table, (2) writes a permanent row to workflow_actions, and (3) triggers a notification to the relevant user.

6.1 Application Stages

Stage	Plain English Meaning
Draft	Applicant started the form but has not submitted yet. No fee paid.
Submitted	Formally submitted with payment. Now in Nodal Officer A's bin
WithNodalOfficerA	In Nodal officer's bin to perform an action
WithTechnicalOfficer	In Technical Officer's bin to perform an action
QuerySent	Technical Officer has sent a query. Applicant has 30 days to respond.
WithExpertCommittee	Application cleared scrutiny. Expert Committee is now evaluating it.
WithNodalPointB	EC has decided. Nodal Point B is uploading the approved EC decision
DecisionPending	EC decision uploaded. Technical Officer is preparing the decision letter.
Approved	Approval letter issued. Approval number generated. Application is CLOSED successfully.
Rejected	Rejection letter issued. Applicant may file an Appeal within allowed period.
WithCEO	Applicant filed an Appeal. CEO is reviewing it.
WithChairperson	CEO rejected Appeal. Applicant filed a Review. Chairperson is evaluating. FINAL STAGE.
Withdrawn	Applicant or authorities withdrew the application. Permanently closed.
Closed	Application resolved through all stages and formally closed.

6.2 Main Application Workflow Diagram

This is the standard flow for NSF, CA, AA, and Any Other applications. rPET has a modified flow described in Section 6.5.

APPLICANT: Selects application type (NSF / CA / AA / rPET / Any Other)

APPLICANT: Fills multi-step form, uploads documents, makes payment

| Stage changes to: Submitted

NODAL OFFICER A: Application appears in their queue

| Assigns to a Technical Officer

TECHNICAL OFFICER: Performs document scrutiny — reviews all uploaded files and compliance information

PATH A — Query Needed TO drafts query → Nodal Officer A approves and sends → Applicant receives query → Applicant has 30 days to respond → Response returned → TO reviews response → If satisfied, recommends EC

PATH B — No Query Required – TO recommends to EC -Application lands in EC Bin.

| Stage: WithExpertCommittee

EXPERT COMMITTEE: Evaluates the full application and records their decision

| Decision recorded → NODAL POINT B uploads approved EC minutes

TECHNICAL OFFICER: Prepares formal decision letter (Approval / Rejection / Clarification)

| Letter sent to Nodal Officer A

NODAL OFFICER A: Sends decision letter to Applicant

APPROVED Approval number generated. Application closed successfully. Applicant sees it in 'Approval Issued' dashboard.

CLARIFICATION Application returned to applicant with queries. Applicant has 30 +15 days to respond.

REJECTED Formal rejection issued. Applicant may file an Appeal within the allowed period.

6.3 Appeal Workflow Diagram

Triggered only after a Rejection decision. The original application record remains; appeals are tracked in the appeals table.

APPLICANT receives REJECTION → Files APPEAL from their dashboard with written grounds and supporting documents

| appeal record created in appeals table

CEO: Reviews full application history + rejection reason + applicant's appeal grounds

CEO APPROVES APPEAL Application restored to active processing. Moves to Nodal Officer's bin as directed by CEO. Final outcome will be Approved.

CEO REJECTS APPEAL Applicant notified. Applicant may now file a REVIEW to the Chairperson as final escalation.

| If CEO rejected → Applicant files REVIEW

CHAIRPERSON: Reviews full history including original rejection, CEO rejection, and applicant's review grounds

CHAIRPERSON APPROVES Application closed as Approved. This is final.

CHAIRPERSON REJECTS Application permanently closed as Rejected. No further escalation possible under any circumstance.

6.4 Query Loop Diagram

| A single application can go through this loop multiple times if responses are insufficient.

Technical Officer identifies that more information is needed from the applicant

|

Technical Officer drafts the query (query record created with status: Draft)

| Sent to Nodal Officer A for approval

Nodal Officer A reviews and approves the query

| Query dispatched — status changes to: Sent — 30-day clock starts

APPLICANT: Views query in 'Reverted Applications' dashboard. Has 30 days to respond.

APPLICANT RESPONDS Response submitted with supporting documents. query.status = Responded. Returns to Nodal Officer A then to Technical Officer.

APPLICANT REQUESTS TIME EXTENSION Applicant cannot meet deadline. Files time extension request. Nodal Officer / TO approves or rejects. New deadline set if approved.

| Technical Officer reviews the response

RESPONSE SUFFICIENT Technical Officer satisfied. Recommends application for Expert Committee. Workflow moves forward.

MORE CLARIFICATION REQUIRED Technical Officer raises another query. The entire loop repeats from the top.

6.5 rPET Workflow Differences

rPET (recycled plastic packaging) follows a modified version of the standard workflow:

Standard Flow	rPET Flow
Application Type: NSF / CA / AA	Application Type: rPET — Fresh / Annual Audit / Conditionally Authorized
Outcome: Approved / Rejected	Outcome: Authorized / Not Authorized / Provisionally Authorized
No inspection step	Mandatory pre-authorization site inspection of the manufacturing unit
One-time approval	Annual Audit required — authorized manufacturers re-audited each year
Appeal to CEO possible	Different dispute path — no standard Appeal/Review section
Nodal: 'Granted Approval'	Nodal: 'Granted Authorization'
Dashboard: 'Approval Issued'	Dashboard: 'Authorized to Manufacture rPET'

6.6 Stage Transition Rules

The backend workflow.service.js file enforces these rules. An application can ONLY move between these pairs — any other transition is rejected:

From Stage	To Stage	Who Triggers It	Action Type
Draft	Submitted	Applicant	Submit with payment
Submitted	WithNodalOfficerA	System (automatic)	Application enters queue
WithNodalOfficerA	WithTechnicalOfficer	Nodal Officer A	Assign to TO
WithTechnicalOfficer	QuerySent	Technical Officer + Nodal Officer A	Draft query → approve → send
QuerySent	WithTechnicalOfficer	Applicant	Submit query response
WithTechnicalOfficer	WithExpertCommittee	Technical Officer + Nodal Officer A	Recommend to EC → approve
WithExpertCommittee	WithNodalPointB	Expert Committee	Record EC decision
WithNodalPointB	DecisionPending	Nodal Point B	Upload EC minutes
DecisionPending	Approved / Rejected	Nodal Officer A	Dispatch decision letter
Rejected	WithCEO	Applicant	File appeal
WithCEO	Approved / Closed	CEO	CEO approves appeal

WithCEO	WithChairperson	Applicant	File review after CEO rejects
WithChairperson	Approved / Closed	Chairperson	Final decision - END
Any stage	Withdrawn	Applicant / Nodal Officer A	Withdrawal request processed

7. API Reference

All frontend-to-backend communication is via REST API. Base URL: <http://localhost:5000/api> (development). All responses are JSON format.

Auth column — Public: no login needed. JWT: any logged-in user. Role name: only that specific role can call this endpoint.

7.1 Authentication Endpoints

Method	Endpoint	Auth	Description
POST	/auth/login/applicant	Public	Applicant login via FSSAI license number + password. Returns JWT token.
POST	/auth/login/authority	Public	Authority login via Username + password. Returns JWT with role embedded.
POST	/auth/register	Public	New applicant registration. Creates user + applicant_profile records.
POST	/auth/logout	JWT	Invalidates the current session token.
POST	/auth/forgot-password	Public	Sends password reset link to registered email.
POST	/auth/reset-password	Public	Resets password using token from email link.

7.2 Application Endpoints

Method	Endpoint	Auth	Description
GET	/applications	JWT	List applications. Auto-filtered by logged-in user's role — applicants can see their applications; officers can see the applications in their bin.
POST	/applications	Applicant	Create new application (saved as Draft).
GET	/applications/:id	JWT	Full application detail. Role determines visible fields and available actions.
PATCH	/applications/:id	Applicant	Update a draft before submission.

POST	/applications/:id/submit	Applicant	Formally submit (records payment, generates application reference number, changes stage to Submitted).
POST	/applications/:id/withdraw	Applicant	Withdraw a submitted application.
GET	/applications/:id/history	JWT	Full workflow audit trail for this application from workflow_actions table.
POST	/applications/:id/documents	Applicant	Upload a document to an application.
GET	/applications/:id/documents	JWT	List all documents for this application.

7.3 Workflow Action Endpoints

Method	Endpoint	Auth	Description
POST	/workflow/assign	NodalOfficerA	Assign application to a specific Technical Officer.
POST	/workflow/approve-query	NodalOfficerA	Approve a drafted query and send to the applicant.
POST	/workflow/forward-to-ec	NodalOfficerA	Forward application to Expert Committee (after TO recommendation).
POST	/workflow/dispatch-decision	NodalOfficerA	Send prepared decision letter to applicant.
POST	/workflow/draft-query	TechnicalOfficer	Draft a query for Nodal Officer A to review and approve.
POST	/workflow/recommend-ec	TechnicalOfficer	Recommend application for Expert Committee forwarding.
POST	/workflow/prepare-letter	TechnicalOfficer	Save draft of decision letter (approval/rejection/clarification).
POST	/workflow/respond-to-query	Applicant	Submit response to an active query.
POST	/workflow/record-ec-decision	ExpertCommittee	Record the EC's decision outcome.
POST	/workflow/upload-ec-decision	NodalPointB	Upload approved EC decision and confirm the decision in the system.

7.4 Appeal, Review, and Extension Endpoints

Method	Endpoint	Auth	Description
POST	/appeals	Applicant	File a new appeal against a rejection.
GET	/appeals/:id	JWT	Get appeal detail including CEO decision if made.
POST	/appeals/:id/decide	CEO	Record CEO's decision on an appeal (Approved or Rejected).
POST	/reviews	Applicant	File a review petition after CEO rejects an appeal.
GET	/reviews/:id	JWT	Get review detail.
POST	/reviews/:id/decide	Chairperson	Record Chairperson's FINAL decision on a review.
POST	/extensions	Applicant	Request a time extension for a specific query.
PATCH	/extensions/:id	NodalOfficerA	Approve or reject a time extension request.

7.5 Reports and Admin Endpoints

Method	Endpoint	Auth	Description
GET	/reports/status	NA or TO	Status sheet with ageing buckets — authority-side and applicant-side.
GET	/reports/search	NA or TO	Full lifecycle search with all filters (company, dates, status, type, etc.).
GET	/reports/approved	NA or TO	All approved applications report.
GET	/reports/rejected	NO or TO	All rejected applications report.
GET	/notifications	JWT	Notifications list for logged-in user.
PATCH	/notifications/:id/read	JWT	Mark notification as read.
GET	/admin/applications	Admin	Read-only list of ALL applications with current officer/stage.
GET	/admin/officers	Admin	All officers grouped by role.
POST	/admin/officers	Admin	Create a new officer account.
POST	/admin/assignments	Admin	Assign an officer to an application or application type.

GET	/admin/tracking/:id	Admin	Full tracking view — which officer holds an application and at what stage.
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8. Reference Number Generation

Two auto-generated numbers are created by the backend's numbering service — one when an application is submitted, and one when it is approved.

8.1 Application Reference Number

Format: YEAR - MONTH - AppTypeCode - CategoryCode - SerialNo

Component	Example	Notes
YEAR	2026	4-digit year
MONTH	04	2-digit month
AppTypeCode	NSF	Configurable in database — NSF CA AA RPET OTH
CategoryCode	FOOD	FoSCoS category code — configurable
SerialNo	0001	Auto-incremented, 4 digits, resets per year
Full Example	2026-04-NSF-FOOD-0001	What the applicant sees as their reference

8.2 Approval Number

Format: YEAR - DecisionCode - AuthorityCode - CategoryCode - ApprovalID

Component	Example	Notes
YEAR	2026	4-digit year
DecisionCode	APP	APP for Approved AUTH for Authorized (rPET)
AuthorityCode	FSSAI	FSSAI office code — configurable
CategoryCode	NSF	Same as application category code
ApprovalID	00042	Unique auto-incremented approval sequence
Full Example	2026-APP-FSSAI-NSF-00042	Printed on the official approval letter (Form II)

All code values are stored in the fee_config and application_type_codes configuration tables. The Admin can change them without any code modifications.

9. Notification System

The Notification Layer is a standalone service (separate from the main backend) that sends Email and SMS alerts. The Application Layer calls it after every workflow event. Notifications are also stored in the notifications database table for in-app display.

9.1 Notification Events

Event	Who Gets Notified	Channels
Application submitted successfully	Applicant	Email + SMS + In-App
Application assigned to Technical Officer	Technical Officer	Email + In-App
Query approved and dispatched to applicant	Applicant	Email + SMS + In-App
Applicant submits query response	Technical Officer + Nodal Officer A	Email + In-App
Application forwarded to Expert Committee	EC Members	Email + In-App
EC decision uploaded by Nodal Point B	Technical Officer + Nodal Officer A	In-App
Decision letter dispatched to applicant	Applicant	Email + SMS + In-App
Application APPROVED — approval number issued	Applicant	Email + SMS + In-App
Application REJECTED	Applicant	Email + SMS + In-App
Appeal filed by applicant	CEO + Nodal Officer A	Email + In-App
CEO decision on appeal	Applicant	Email + In-App
Review filed by applicant	Chairperson + Nodal Officer A	Email + In-App
Chairperson final decision	Applicant	Email + SMS + In-App
Time extension requested	Nodal Officer A + Technical Officer	In-App
Time extension approved or rejected	Applicant	Email + In-App
Query deadline approaching (3 days remaining)	Applicant	Email + SMS
Query deadline expired without response	Nodal Officer A	In-App

9.2 Notification Layer Folder Structure

File / Folder	Purpose
notification-layer/templates/	HTML email templates — one file per event type
notification-layer/providers/email.provider.js	Nodemailer wrapper — sends emails
notification-layer/providers/sms.provider.js	MSG91 / Twilio wrapper — sends SMS
notification-layer/notification.service.js	Main entry point — called by application-layer after each workflow event
notification-layer/notification.queue.js	Bull + Redis queue — processes notifications asynchronously so API does not wait

10. Security

Control	Implementation	Why It Matters
Authentication	JWT tokens signed with secret key, expire after 8 hours	Only logged-in users access protected data
Role-based access control	Every API endpoint checks role from JWT before allowing action	An applicant cannot perform officer actions; officers cannot cross role boundaries
Password storage	bcryptjs hashing — passwords never stored as plain text	Database compromise cannot expose user passwords
HTTPS in production	All traffic encrypted in transit	Data interception prevented
Security headers	Helmet.js middleware	Prevents XSS, clickjacking, and other common web attacks
CORS policy	Only the approved frontend domain can call the API	Unauthorized websites cannot make API calls
Rate limiting	200 requests/minute globally; 10 requests/minute for login	Prevents brute-force and denial-of-service attacks
Input validation	All incoming data validated before processing	Malformed or malicious data rejected before reaching business logic
SQL injection prevention	All DB queries use parameterized statements	Attackers cannot manipulate database queries via user input
File upload validation	File type and size checked before storing	Malicious files blocked at upload point
Audit trail	Every action permanently recorded in workflow_actions with actor and timestamp	Full accountability and forensic investigation capability
Admin read-only enforcement	Admin role middleware blocks all write operations at API level — not just frontend	Even if Admin frontend is bypassed, the API refuses workflow actions

11. Reports and Ageing

The application provides ageing buckets — predefined time range groupings — so overdue applications are immediately visible.

11.1 Authority-Side Ageing

Bucket	Time Range	Status
On Time	1 to 45 days pending with authority	Within acceptable processing window
Delayed	46 to 75 days pending with authority	Requires management attention
Critical / Long Outstanding	More than 75 days	Seriously overdue — escalation needed

11.2 Applicant-Side Ageing (Query Response)

Bucket	Time Range	Status
On Time	1 to 30 days pending with applicant	Within the 30-day query response window
Delayed	31 to 45 days	Approaching or past deadline
Critical	More than 45 days	Application Freeze

11.3 Available Reports

Report	Description
Approved Applications	All approved applications with approval numbers, dates, and EC references. Category-wise and year-wise views.
Rejected Applications	All rejected applications with rejection dates and grounds.
Withdrawn Applications	Applications withdrawn by applicant or closed by FSSAI.
Appeal Applications	All applications where an appeal was filed — includes CEO decision and date.
Review Applications	All applications where a review petition was filed — includes Chairperson decision.
Status Report (Ageing)	Cross-tabulation: applications by stage, type, and ageing bucket. Two sheets: summary and detailed counts.
Track Application / Approved	Search by reference number or approval number to get the full lifecycle of one application.
Search Console	Full multi-filter search across all applications using any combination of: company, product, category, status, date ranges, type, reference number.

12. Project Folder Structure

The project has four top-level folders matching the four architecture layers from SRS Section 2.2. Each layer is independent.

12.1 Root

```
epaas/
├── presentation-layer/    ← React.js frontend
├── application-layer/     ← Node.js + Express.js backend
├── database-layer/       ← PostgreSQL migrations and model query files
└── notification-layer/   ← Email and SMS notification service
```

12.2 Application Layer (Backend) Detail

```
application-layer/
├── server.js              ← Entry point: starts Express server + DB
├── connection
├── app.js                ← Registers all routes, middleware, and error
├── handler
├── .env                  ← Environment variables (never committed to
├── git)                  ← Template with empty values (committed to git)
├── .env.example
├── package.json
├── src/
│   ├── config/
│   │   ├── db.js        ← PostgreSQL connection pool (pg library)
│   │   └── constants.js ← ROLES, APPLICATION_STAGES, ACTION_TYPES
│   ├── routes/          ← URL definitions (one file per feature area)
│   │   ├── auth.routes.js
│   │   ├── application.routes.js
│   │   ├── workflow.routes.js
│   │   ├── query.routes.js
│   │   ├── decision.routes.js
│   │   ├── appeal.routes.js
│   │   ├── review.routes.js
│   │   ├── extension.routes.js
│   │   ├── reports.routes.js
│   │   ├── notification.routes.js
│   │   └── admin.routes.js
│   ├── controllers/     ← Receives request → calls service → sends JSON
│   ├── response
│   │   ├── services/
│   │   │   ├── workflow.service.js ← ALL business logic lives here
│   │   │   │   └── workflow.service.js ← Most important file: enforces all stage
│   │   └── transitions
│   │       ├── numbering.service.js ← Generates reference and approval numbers
│   │       ├── application.service.js
│   │       ├── query.service.js
│   │       ├── decision.service.js
│   │       ├── appeal.service.js
│   │       ├── review.service.js
│   │       ├── reports.service.js
│   │       └── upload.service.js
│   └── middleware/
│       └── auth.middleware.js ← Verifies JWT token on every protected route
```

```
roles | └─ role.middleware.js      ← checkRole('NodalOfficerA') - blocks wrong
      |
      └─ upload.middleware.js  ← multer config for file uploads
          └─ errorHandler.middleware.js
      └─ utils/
          └─ logger.js          ← Request and error logging
          └─ dateHelpers.js     ← Ageing calculations, deadline checks
          └─ referenceNumber.js ← Auto-generates application and approval
numbers └─ mailer.js            ← Calls notification-layer
```

12.3 Database Layer Detail

```
database-layer/
├─ migrations/                                ← Run these in order to build the database
│   └─ 001_create_users.sql
│   └─ 002_create_applicant_profiles.sql
│   └─ 003_create_applications.sql
│   └─ 004_create_application_documents.sql
│   └─ 005_create_workflow_actions.sql
│   └─ 006_create_queries.sql
│   └─ 007_create_decisions.sql
│   └─ 008_create_approval_numbers.sql
│   └─ 009_create_appeals.sql
│   └─ 010_create_reviews.sql
│   └─ 011_create_time_extension_requests.sql
│   └─ 012_create_notifications.sql
│   └─ 013_create_officer_assignments.sql
│   └─ 014_create_fee_config.sql
├─ models/                                    ← Raw SQL query functions (one file per table)
│   └─ user.model.js
│   └─ application.model.js
│   └─ workflow.model.js
│   └─ ... (one per table)
├─ seeds/                                    ← Initial data: roles, fee config, officer
└─ accounts
```

13. Non-Functional Requirements

NFR ID	Category	Requirement
NFR-01	Role Security	Users shall only access records and actions permitted to their specific role. Enforced at API level — not just frontend.
NFR-02	Auditability	All assignments, queries, decisions, appeals, reviews, and extension actions permanently logged with timestamp and actor identity in workflow_actions table.
NFR-03	Searchability	Search console and operational queues shall support fast filtering on all fields defined in Section 11.3 of this document.
NFR-04	Usability	Dashboards present counts, pending actions, and notifications in a simple role-based layout navigable by non-technical government users.
NFR-05	Document Retention	All uploaded documents and decision letters remain permanently linked to application history. Files are never deleted.
NFR-06	Configurability	Application type codes, authority codes, product category codes, and fee amounts are configurable by Admin through database configuration tables — no code changes required.
NFR-07	Performance	Search and report queries return results within 3 seconds for up to 10,000 application records.
NFR-08	Availability	System targets 99% uptime during government working hours (9 AM to 6 PM IST, Monday to Saturday).

14. Application Fee Structure

All fees are stored in the fee_config database table and configurable by Admin. Current schedule as defined in SRS Section 2.4:

#	Application Type / Category	Claim Type	Fee (INR)
1	NSF / E-PAAS (standard)	—	50,000 + GST
2	Ayurveda Aahara — Category A	No label claim selected	No fee
3	Ayurveda Aahara — Category A	Disease Risk Reduction	50,000
4	Ayurveda Aahara — Category A	Health Benefit	7,500
5	Ayurveda Aahara — Category A	No Claim (without label claim)	7,500
6	Ayurveda Aahara — Category B	No label claim selected	No fee
7	Ayurveda Aahara — Category B	Disease Risk Reduction	50,000
8	Ayurveda Aahara — Category B	Health Benefit	10,000
9	Ayurveda Aahara — Category B	No Claim	10,000
10	Ayurveda Aahara — Category B1	No label claim selected	No fee
11	Ayurveda Aahara — Category B1	Disease Risk Reduction	50,000
12	Ayurveda Aahara — Category B1	Health Benefit	10,000
13	Ayurveda Aahara — Category B1	No Claim	10,000
14	Ayurveda Aahara — Category B2	No label claim selected	No fee
15	Ayurveda Aahara — Category B2	Disease Risk Reduction	50,000
16	Ayurveda Aahara — Category B2	Health Benefit	10,000
17	Ayurveda Aahara — Category B2	No Claim	10,000
18	Claim Approval (CA)	—	50,000

15. Glossary

Term	Definition
E-PAAS	Electronic Product and Claim Approval Application System
FSSAI	Food Safety and Standards Authority of India
Applicant	Applicant applying for non-specified food and food ingredients
NSF	Non-Specified Food
CA	Claim Approval — approval for a health/disease claim on a food label
AA	Ayurveda Aahara — Ayurvedic food category with sub-types A, B, B1, B2
rPET	Recycled Polyethylene Terephthalate — recycled plastic used for food packaging; needs authorization
Nodal Officer A	Primary Authorised officer — receives applications, assigns officers, sends decisions
Technical Officer	Authorised officer who scrutinizes documents and drafts queries and letters
Expert Committee	Panel of experts who evaluate applications and record formal decisions
Nodal Point B	Authorised officer who uploads approved EC decisions into E-PAAS
CEO	Chief Executive Officer of FSSAI — appellate authority for rejected applications
Chairperson	Chairperson of FSSAI — final and absolute authority for review petitions
JWT	JSON Web Token — a digital key issued at login that proves who you are to the backend
UUID	Universally Unique Identifier — 36-character random ID used as primary key in all tables
PK	Primary Key — unique identifier column for each row in a table
FK	Foreign Key — a column that references the primary key of another table, creating a relationship
Scrutiny	Detailed technical review of an application's documents and compliance information by Technical Officer
EC	Expert Committee — shorthand used throughout the system
FoSCoS	Food Safety Compliance System
Form II	Approval/ Rejection Decision Template
Stage	The current position of an application in the workflow — determines which officer sees it and what actions are available
Ageing Bucket	A predefined time range used to classify how long an application has been pending at a particular stage
